

## 2.5 Nature of Light

This topic covers the wave/particle nature of light.

This topic may be studied either by using applications that relate to, for example, solar cells or by the historical study of the nature of light.

| Students will be assessed on their ability to:                                                                                                                                                   | Suggested experiments                                            |
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| 63 explain how the behaviour of light can be described in terms of waves and photons                                                                                                             |                                                                  |
| 64 recall that the absorption of a photon can result in the emission of a photoelectron                                                                                                          | Demonstration of discharge of a zinc plate by ultra violet light |
| 65 understand and use the terms <i>threshold frequency</i> and <i>work function</i> and recognise and use the expression $hf = \phi + \frac{1}{2}mv_{max}^2$                                     |                                                                  |
| 66 use the non-SI unit, the electronvolt (eV) to express small energies                                                                                                                          |                                                                  |
| 67 recognise and use the expression $E = hf$ to calculate the highest frequency of radiation that could be emitted in a transition across a known energy band gap or between known energy levels |                                                                  |
| 68 explain atomic line spectra in terms of transitions between discrete energy levels                                                                                                            | Demonstration using gas-filled tubes                             |
| 69 define and use radiation flux as power per unit area                                                                                                                                          |                                                                  |
| 70 recognise and use the expression $\text{efficiency} = \frac{[\text{useful energy (or power) output}]}{[\text{total energy (or power) input}]}$                                                |                                                                  |
| 71 explain how wave and photon models have contributed to the understanding of the nature of light                                                                                               |                                                                  |
| 72 explore how science is used by society to make decisions, for example, the viability of solar cells as a replacement for other energy sources, the uses of remote sensing                     |                                                                  |